



Analytical and Approximate Solution to Fractional Diffusion Models

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Abstract: The scheme of the proposed method, reduced differential transformed method (RDTM), provided the solution of time-dependent fractional reaction-diffusion models. Reaction-diffusion systems often appear in the modeling for an essential basis of the processes of morphogenesis in the field of biology (and may even be observed in skin pigmentation and animal coats), for pattern formation of a prototype model (like spirals, fronts, targets, stripes, hexagons, and dissipative solitons), including the spread of epidemics, ecological invasions, wound healing, and tumor growth. In this study, the behavior and analytical treatment of nonlinear time-dependent fractional Cauchy reaction-diffusion partial differential equations (CRDPDE) have been studied via the scheme of RDTM. The finding was interesting to reveal that the scheme was and consistent for the solutions to the problems concerned. The present study suggests that RDTM is simpler and more easily calculable than other methods, hence applicable to any coupled systems.

Keywords: Fractional Cauchy reaction; Fractional partial differential equation; Reduced differential transformed method; Approximate solution; Fractional differential equation

1 Introduction

In the recent era, fractional calculus has gained much importance in the applied fields of Science and Engineering. One after the other, many researchers and scientists like Leibniz, Bernnoui, Lee-Hopital, Euler, Walis, and Reimann Liouville, have made great contributions to the fractional calculus and its research area. Its contributions and applications are devoted to the fields of damping viscoelasticity, electronics, genetic algorithms, biology, telecommunication, signal processing, traffic systems, robotic technology, finance, chemistry, physics, and economics. Various phenomena in different fields can be modeled by representing partial differential equations (PDEs) of integer or non-integer orders. In this concern, numerous scientists and researchers have used different schemes for the numerical treatment of linear/nonlinear fractional partial differential equations (FPDEs) [1–4]. Many phenomena of reaction-diffusion systems can be expressed in the form of PDEs of integer or non-integer orders. Reaction-diffusion systems often appear in the modeling for an essential basis of the processes to morphogenesis in the field of biology [5, 6] (and may even be observed to skin pigmentation and animal coats [7, 8]), for pattern formation of a prototype model [9] (like spirals, fronts, targets, stripes, hexagons, and dissipative solitons), including the spread of epidemics [10], ecological invasions [11], wound healing [12] and tumor growth [13–15]. As a cloud of particles spreads faster than predicted by the classical equation, it is a useful application of fractional diffusion equations. When we replace the second derivative in the spatial variable with a fractional derivative of order less than two in a fractional diffusion equation, then the obtained solutions spread faster than the classical solutions, which may exhibit asymmetry depending on the fractional derivative used. Therefore, for our current study, we consider fractional Cauchy reaction-diffusion partial differential equations (CRDPDE):

$$\frac{\partial^\beta \omega(r, s)}{\partial s^\beta} = c \frac{\partial^2 \omega(r, s)}{\partial r^2} + r_1(r, s) \omega(r, s), \quad (r, s) \in \Omega \quad (1)$$

When $\beta = 1$ then Eq. (1) convert in the form of classical reaction diffusion equation, where $c \frac{\partial^2 \omega(r, s)}{\partial r^2}$ and $r_1(r, s) \omega(r, s)$ are the diffusion and reaction terms respectively, whereas $\omega(r, s)$ is the concentration, $r_1(r, s)$ is the

reaction parameter, $c(r, s)$ is the diffusion co-efficient, (r, s) are the spatial and time dependent variable of Eq. (1). On the other hand, due to their frequent appearance in numerous applications of fractional differential equations (FDEs), some research articles have studies of a few aspects of FDEs, such as existence and uniqueness of solutions of Cauchy-type problems [16]. Among the important topics, the research seeks numerical solutions as well as exact solutions of FDEs. Many powerful and efficient techniques have been developed in the past to find exact and numerical solutions to FDEs [4, 17]. We have investigated and studied the scheme of proposed method, named the reduced differential transformed method (RDTM) [18–20] for the solution of the CRDPDE. The disclosed finding is very interesting. For testing of concern method, the results are provided in distinct tables and figures, which reveal that scheme is effective and consistent for the solution of the concerned problems. However, for the numerical treatment of any type of differential equation, one can choose one of the proposed methods by observing the data available in the numerical sections of this work. The recommendation of the work is disposed in the coming sections as: Section 2 is the basic definitions and main idea of RDTM, whereas some test problems have been provided in Section 3. A brief conclusion of this work is given in Section 4.

2 Traditional Differential Transform Method

2.1 The Differential Transformation Technique of One Dimensional Function

Let $\omega(r)$ be a function, then differential transformation of $\omega(r)$ is given as follows [4]:

$$W(k) = \frac{1}{k!} \left[\frac{\partial^k \omega(r)}{\partial r^k} \right]_{r=0} \quad (2)$$

where, $W(k)$ is the transformed function and $\omega(r)$ is the original function. Likewise, $\frac{\partial^k}{\partial r^k}$ represents the order k derivative with respect to x . But inverse transformation of $W(k)$ can be obtained as

$$\omega(r) = \sum_{k=0}^{\infty} W(k) r^k \quad (3)$$

By using the Eq. (2) and Eq. (3) we get

$$\omega(r) = \sum_{k=0}^{\infty} \frac{1}{k!} \left[\frac{\partial^k \omega(r)}{\partial r^k} \right]_{r=0} r^k \quad (4)$$

In the view of above mentioned definitions it is clear that the concept of differential transformation is derived from the expansion of Taylor series. Moreover, Eq. (2) and Eq. (3) represents the basic mathematical operations, which are given in Table 1.

Table 1. Differential transformation of one dimensional functions

Serial Number	Function Form	Differential Transformed Form
1	$v(r) = \omega(r) \pm \mu(r)$	$V(k) = W(k) \pm U(k)$
2	$v(r) = c\omega(r)$	$W(k) = cU(k)$
3	$v(r) = \frac{\partial^j \omega(r)}{\partial r^j}$	$V(k) = \frac{(k+j)!}{k!} W(k+j)$
4	$v(r) = \omega(r)\mu(r)$	$V(k) = \sum_{i=0}^j W(i)U(j-i)$

2.2 Differential Transformation Technique of Two Dimensional Functions

Similarly, the differential transformation of two dimensional of a function $\omega(r, s)$ can be obtained as [7, 8]

$$W(k, h) = \frac{1}{k!h!} \left[\frac{\partial^{k+h}}{\partial x^k \partial t^h} \omega(x, t) \right]_{(0,0)} \quad (5)$$

where, $\omega(r, s)$ and $W(k, h)$ is the original function and transformed function respectively. But the differential inverse transform of $W(k, h)$ is provided as

$$\omega(r, s) = \sum_{k=0}^{\infty} \sum_{h=0}^{\infty} W(k, h) r^k s^h \quad (6)$$

Then combining Eq. (5) and Eq. (6), we write

$$\omega(r, s) = \sum_{k=0}^{\infty} \sum_{h=0}^{\infty} \frac{1}{k!h!} \left[\frac{\partial^{k+h}}{\partial x^k \partial t^h} \omega(r, s) \right]_{(0,0)} r^k s^h \quad (7)$$

The mathematical operations of differential transformation of two dimensional functions shown in Table 2.

Table 2. Differential transformation of two dimensional functions

Serial Number	Function Form	Differential Transformed Form
1	$v(r, s) = \omega(r, s) \pm \mu(r, s)$	$V(k, h) = W(k, h) \pm U(k, h)$
2	$v(r, s) = c \omega(r, s)$	$V(k, h) = c W(k, h)$
3	$v(r, s) = \frac{\partial \omega(r, s)}{\partial r}$	$V(k, h) = (k+1)W(k+1, h)$
4	$v(r, s) = \frac{\partial \omega(r, s)}{\partial s}$	$V(k, h) = (h+1)W(k, h+1)$
5	$v(r, s) = \frac{\partial^{p+q} \omega(r, s)}{\partial r^p \partial s^q}$	$V(k, h) = \frac{(k+p)!(h+q)!W(k+p, h+q)}{k!q!}$
6	$v(r, s) = \omega(r, s)\mu(r, s)$	$V(k, h) = \sum_{i=0}^k \sum_{j=0}^h W(i, h-j)U(k-i, j)$

The basic operations of RDTM for the purpose to explain and know the idea of RDTM are follows in Table 3.

Table 3. Reduced differential transformation of the functions

Serial Number	Function Form	Transformed Form
1	$v(r, s)$	$V(r, s) = \frac{1}{k!} \left[\frac{\partial^k v(r, s)}{\partial s^k} \right]_{s=0}$
2	$v(r, s) = c$, where c is a constant	$V_k(r) = \delta(k) = \begin{cases} 1 & \text{if } k = 0 \\ 0 & \text{if } k \neq 0 \end{cases}$
3	$v(r, s) = \mu(r, s) + \omega(r, s)$	$V_k(r) = U_k(r) + W_k(r)$
4	$v(r, s) = c\mu(r, s)$	$V_k(r) = cU_k(r)$
5	$v(r, s) = \mu(r, s)\omega(r, s)$	$V_k(r) = \sum_{l=0}^k U_l(r)W_{k-l}(r)$
6	$v(r, s) = r^m s^n$	$V_k(r) = r^m \delta(k-n) = \begin{cases} r^m & \text{if } k = n \\ 0 & \text{if } k \neq n \end{cases}$
7	$v(r, s) = r^m s^n \mu(r, s)$	$V_k(r) = r^m U_{k-n}(r)$
8	$v(r, s) = \frac{\partial}{\partial s} \mu(r, s)$	$V_k(r) = (k+1)U_{k+1}(r)$
9	$v(r, s) = \frac{\partial^m}{\partial r^m} \mu(r, s)$	$V_k(r) = \frac{\partial^m}{\partial r^m} U_k(r)$

3 Test Problems

The illustration of RDTM is our concern in this section of the manuscript. We shall apply RDTM to obtain the approximate solutions of CRDPDE.

Problem 1: Consider

$$\frac{\partial^\beta \omega(r, s)}{\partial s^\beta} = \frac{\partial^2 \omega(r, s)}{\partial r^2} - \omega(r, s) \quad s > 0, 0 < \beta \leq 1 \quad (8)$$

Condition is,

$$\omega(r, 0) = e^{-r} + r,$$

the above equation can be written in reduced differential transform form as

$$\frac{\Gamma(1 + (k+1)\beta)}{\Gamma(1 + k\beta)} W_{k+1}(r) = \frac{\partial^2 W_k(r)}{\partial r^2} - W_k(r), \quad W_0(r) = e^{-r} + r,$$

or

$$W_{k+1}(r) = \frac{\Gamma(1+k\beta)}{\Gamma(1+(k+1)\beta)} \left(\frac{\partial^2 W_k(r)}{\partial r^2} - W_k(r) \right), \quad W_0(r) = e^{-r} + r.$$

where, $k = 0, 1, 2, 3, \dots$

For $k = 0$,

$$W_1(r) = \frac{1}{\Gamma(1+\beta)} \left(\frac{\partial^2 W_0(r)}{\partial r^2} - W_0(r) \right),$$

with $W_0(r) = e^{-r} + r$, we get

$$W_1(r) = \frac{-r}{\Gamma(1+\beta)}.$$

And for $k = 1$,

$$W_2(r) = \frac{\Gamma(1+\beta)}{\Gamma(1+2\beta)} \left(\frac{\partial^2 W_1(r)}{\partial r^2} - W_1(r) \right),$$

using $W_1(r) = \frac{-r}{\Gamma(1+\beta)}$, we get

$$W_2(r) = \frac{r}{\Gamma(1+2\beta)}.$$

For $k = 2$,

$$W_3(r) = \frac{\Gamma(1+2\beta)}{\Gamma(1+3\beta)} \left(\frac{\partial^2 W_2(r)}{\partial r^2} - W_2(r) \right),$$

using $W_2(r) = \frac{r}{\Gamma(1+2\beta)}$, we get

$$W_3(r) = \frac{-r}{\Gamma(1+3\beta)}.$$

Now using the Taylor series $\tilde{\omega}(r, s) = \sum_{k=0}^{\infty} W_k(r) s^{k\beta}$, by putting the values we get the solution as

$$\tilde{\omega}(r, s) = e^{-r} + r \left(1 - \frac{s^\beta}{\Gamma(\beta+1)} + \frac{s^{2\beta}}{\Gamma(2\beta+1)} - \frac{s^{3\beta}}{\Gamma(3\beta+1)} + \dots \right),$$

$$\tilde{\omega}(r, s) = e^{-r} + r E_\beta(-s^\beta).$$

We substitute $\beta = 1$, then the solution transferred to exact solution of problem 1, which shows that RDTM gives best results. Table 4 indicates RDTM solution, whereas Table 5 shows absolute errors between exact and RDTM solution of problem 1, where these two results (Exact and RDTM) are closed to each other.

Table 4. Reduced differential transformed method (RDTM) solution at $r = 1$ and $\beta = 0.7, 0.8, 0.9$ for some values of s of problem 1

s	RDTM ($\beta = 0.7$)	RDTM ($\beta = 0.8$)	RDTM ($\beta = 0.9$)
0	1.367879441170	1.367879441171	1.367879441170
0.040	1.260637853220	1.290035406321	1.31227343380
0.080	1.201403428890	1.237085406911	1.26687138070
0.120	1.154982620730	1.192583551391	1.22601854580
0.160	1.116061603950	1.153527380330	1.188484887070
0.200	1.08231664250	1.118504709290	1.153641859890
0.240	1.052450540080	1.086679768070	1.121089820791
0.280	1.025640402040	1.057494884520	1.090544183312
0.320	1.001320746220	1.03054917010	1.061787764212
0.360	0.9790809957220	1.005539431110	1.034646976992
0.400	0.9586108001170	0.9822277755220	1.00897846821
0.440	0.9396682446640	0.9604222918710	0.9846609618071
0.480	0.9220601296460	0.9399647696820	0.9615899569071

Table 5. Error of exact and reduced differential transformed method (RDTM) solution at $r = 1$ by taking different values of s and $\beta = 1$ of problem 1

s	Exact Solution ($\beta = 1$)	RDTM Solution ($\beta = 1$)	Error
0	1.367879441171	1.367879441170	0
0.040	1.328668880320	1.328668880320	0
0.080	1.290995787560	1.290995787561	0
0.120	1.254799877892	1.254799877891	0
0.160	1.220023230142	1.220023230141	0
0.200	1.186610194252	1.186610194251	0
0.240	1.154507302242	1.154507302241	0
0.280	1.123663182631	1.123663182631	$1.04083409e^{-17}$
0.320	1.094028478251	1.094028478250	$5.89805982e^{-17}$
0.360	1.065555767241	1.065555767240	$2.67147415e^{-16}$
0.400	1.038199487212	1.038199487210	$1.0477730e^{-15}$
0.440	1.011915862251	1.011915862250	$3.60822483e^{-15}$
0.480	0.9866628329781	0.9866628329780	$1.114907e^{-14}$

In this portion some figures are given for the efficiency of the concerned technique (RDTM). According to Figure 1, when β varies from 0.7 to 1 then RDTM solution closed to exact solution of problem 1, whereas Figure 2 shows RDTM solution for different values of β , but for $\beta = 1$ RDTM solution converge to exact solution are also mentioned in Figure 3.

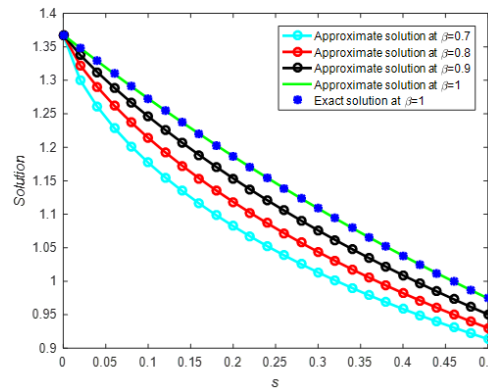


Figure 1. The exact and reduced differential transformed method (RDTM) solutions of problem 1 at $r = 1$ for different values of s and β

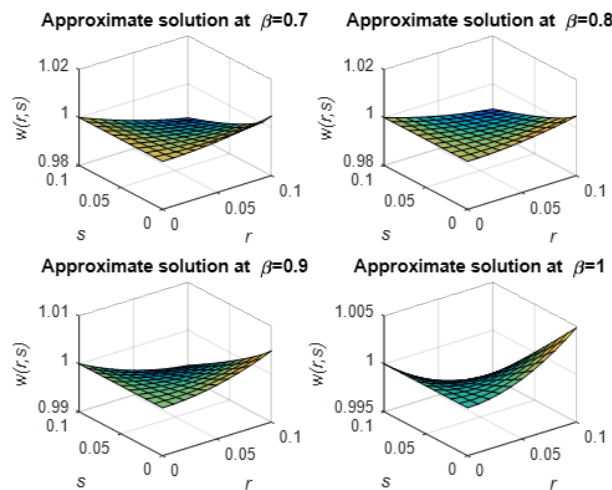


Figure 2. Exact and reduced differential transformed method (RDTM) solutions of problem 1 for different values of r , s and β

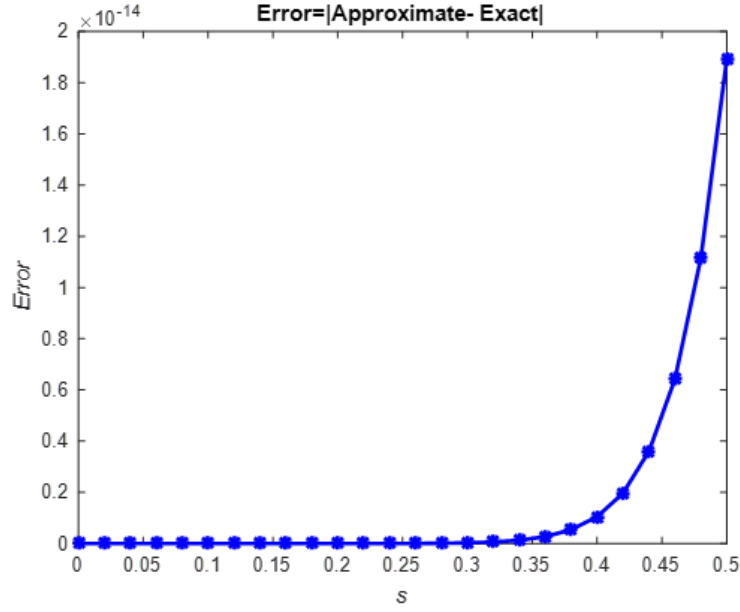


Figure 3. Error between exact and reduced differential transformed method (RDTM) solutions of problem 1 at $r = 1$ for different values of s and $\beta = 1$

Problem 2: Consider

$$\frac{\partial^\beta \omega(r, s)}{\partial s^\beta} = \frac{\partial^2 \omega(r, s)}{\partial r^2} - (1 + 4r^2) \omega(r, s),$$

with initial condition $\omega_0(r, s) = e^{r^2}$, by using reduced differential transform method the above equation can be written as

$$W_{k+1}(r) = \frac{\Gamma(1 + k\beta)}{\Gamma(1 + (k+1)\beta)} \left(\frac{\partial^2 W_k(r)}{\partial r^2} - (1 + 4r^2) W_k(r) \right),$$

with initial $W_0(r) = e^{r^2}$. Where, $k = 0, 1, 2, 3, \dots$

For $k=0$,

$$W_1(r) = \frac{1}{\Gamma(1 + \beta)} \left(\frac{\partial^2 W_0(r)}{\partial r^2} - (1 + 4r^2) W_0(r) \right),$$

with $W_0(r) = e^{r^2}$, we get

$$W_1(r) = \frac{e^{r^2}}{\Gamma(1 + \beta)}.$$

For $k = 1$,

$$W_2(r) = \frac{\Gamma(1 + \beta)}{\Gamma(1 + 2\beta)} \left(\frac{\partial^2 W_1(r)}{\partial r^2} - (1 + 4r^2) W_1(r) \right),$$

with $W_1(r) = \frac{e^{r^2}}{\Gamma(1 + \beta)}$, we get

$$W_2(r) = \frac{e^{r^2}}{\Gamma(1 + 2\beta)}.$$

For $k = 2$,

$$W_3(r) = \frac{\Gamma(1 + 2\beta)}{\Gamma(1 + 3\beta)} \left(\frac{\partial^2 W_2(r)}{\partial r^2} - (1 + 4r^2) W_2(r) \right),$$

with $W_2(r) = \frac{e^{r^2}}{\Gamma(1 + 2\beta)}$, we get

$$W_3(r) = \frac{e^{r^2}}{\Gamma(1+3\beta)}.$$

Now using the Taylor Series $\tilde{\omega}(r, s) = \sum_{k=0}^{\infty} W_k(r) s^{k\beta}$, by putting the values we get the series solution as

$$\tilde{\omega}(r, s) = e^{r^2} \left(1 + \frac{s^\beta}{\Gamma(\beta+1)} + \frac{s^{2\beta}}{\Gamma(2\beta+1)} + \frac{s^{3\beta}}{\Gamma(3\beta+1)} + \dots \right).$$

When substitute $\beta = 1$ to obtain

$$\tilde{\omega}(r, s) = e^{r^2+s},$$

that is the exact solution to integer order CRPDEs.

For problem 2, the results in Table 6 show the comparison of approximate (RDTM) solution for distinct values of β , while the results in Table 7 show absolute error of exact solution and approximate solution obtained by RDTM. The comparison and absolute error of approximate solutions obtained by RDTM show the authentic accuracy of the proposed method.

Table 6. Approximate solution taking $r = 1$ and $\beta = 0.7, 0.8, 0.9$ for different values of s of problem 2

s	RDTM ($\beta = 0.7$)	RDTM ($\beta = 0.8$)	RDTM ($\beta = 0.9$)
0	2.718281828461	2.718281828461	2.718281828461
0.040	3.058244971610	2.951956915421	2.879315539471
0.080	3.299285476061	3.140875300590	3.027290139911
0.120	3.524980511861	3.323457441130	3.175455827511
0.160	3.746157016380	3.505576341110	3.32624969361
0.200	3.967318613220	3.689812629222	3.480850012331
0.240	4.190952462250	3.877663265592	3.640008217330
0.280	4.418673676970	4.070148258322	3.804285974970
0.320	4.651654131110	4.268043114952	3.974148383830
0.360	4.890816949900	4.471986113441	4.150007910350
0.400	5.136936548630	4.682534292591	4.332247778140
0.440	5.39069497830	4.900195411711	4.521235589491
0.480	5.652715954060	5.125447504291	4.717331845071

Note: RDTM = reduced differential transformed method.

Table 7. Errors between exact and reduced differential transformed method (RDTM) solution at $r = 1$ for various values of s and $\beta = 1$ of problem 2

s	Exact Solution ($\beta = 1$)	RDTM Solution ($\beta = 1$)	Error
0	2.71828182846	2.71828182846	0
0.040	2.82921701435	2.82921701435	$6.938893904e^{-18}$
0.080	2.94467955107	2.94467955107	$6.938893904e^{-18}$
0.120	3.06485420329	3.06485420329	0
0.160	3.18993327612	3.18993327612	$6.938893904e^{-18}$
0.200	3.32011692274	3.32011692274	$6.938893904e^{-18}$
0.240	3.45561346476	3.45561346476	$1.387778781e^{-17}$
0.280	3.59663972557	3.59663972557	$2.775557562e^{-17}$
0.320	3.74342137726	3.74342137726	$1.665334537e^{-16}$
0.360	3.8961933018	3.8961933018	$7.702172233e^{-16}$
0.400	4.05519996684	4.05519996684	$3.025357740e^{-15}$
0.440	4.220695817	4.220695817	$1.0422218644e^{-14}$
0.480	4.39294568092	4.39294568092	$3.2446267895e^{-14}$

For justification of RDTM a few figures are given below to clarify the efficiency of the concerned technique (RDTM). In Figure 4, when β varies from 0.7 to 1 then RDTM solution closed to exact solution of problem 2,

whereas Figure 5 shows RDTM solution for different values of β , but for $\beta = 1$ RDTM solution converge to exact solution are mentioned in Figure 6.

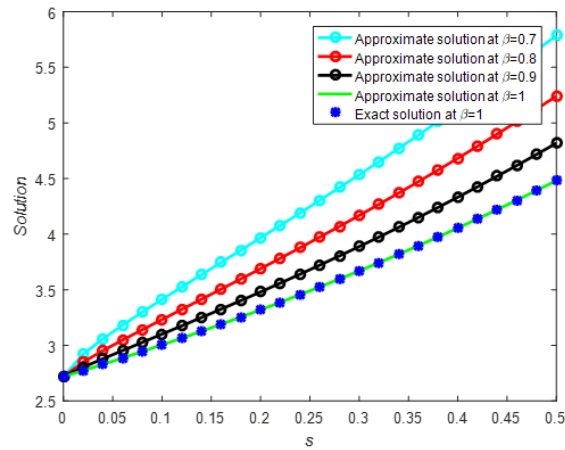


Figure 4. The exact and reduced differential transformed method (RDTM) results of the problem 2 at $r = 1$, taking various values of s and β

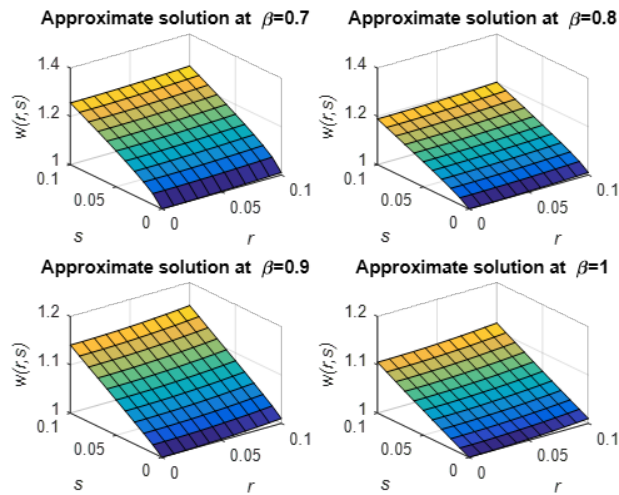


Figure 5. Reduced differential transformed method (RDTM) results of the problem 2 at $r = 1$ for various values of r , s and β

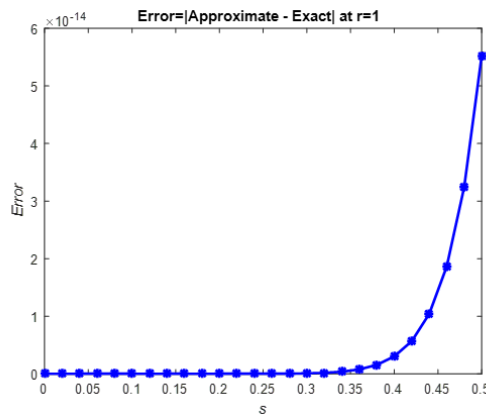


Figure 6. The absolute error of exact and reduced differential transformed method (RDTM) results of the problem 2 at $r = 1$, taking various values of s and $\beta = 1$

Problem 3: Consider

$$\frac{\partial^\beta \omega(r, s)}{\partial s^\beta} = \frac{\partial^2 \omega(r, s)}{\partial r^2} - (1 + 4r^2 - 2s) \omega(r, s),$$

with initial condition $\omega(r, 0) = e^{r^2}$.

We can write the differential equation in reduced differential transform form as

$$W_{k+1}(r) = \frac{\Gamma(1 + k\beta)}{\Gamma(1 + (k+1)\beta)} \left(\frac{\partial^2 W_k(r)}{\partial r^2} - (1 + 4r^2) W_k(r) + 2W_{k-1}(r) \right), W_0(r) = e^{r^2}, W_1(r) = 0,$$

where, $k = 1, 2, 3, \dots$

For $k = 1$,

$$W_2(r) = \frac{\Gamma(1 + \beta)}{\Gamma(1 + 2\beta)} \left(\frac{\partial^2 W_1(r)}{\partial r^2} - (1 + 4r^2) W_1(r) + 2W_0(r) \right),$$

using initial condition $W_0(r) = e^{r^2}$, $W_1(r) = 0$, we get

$$W_2(r) = \frac{2\Gamma(1 + \beta)e^{r^2}}{\Gamma(1 + 2\beta)}.$$

For $k = 2$,

$$W_3(r) = \frac{\Gamma(1 + 2\beta)}{\Gamma(1 + 3\beta)} \left(\frac{\partial^2 W_2(r)}{\partial r^2} - (1 + 4r^2) W_2(r) + 2W_1(r) \right),$$

with $W_2(r) = \frac{2\Gamma(1+\beta)e^{r^2}}{\Gamma(1+2\beta)}$, we get

$$W_3(r) = 0.$$

For $k = 3$,

$$W_4(r) = \frac{\Gamma(1 + 3\beta)}{\Gamma(1 + 4\beta)} \left(\frac{\partial^2 W_3(r)}{\partial r^2} - (1 + 4r^2) W_3(r) + 2W_2(r) \right),$$

with $W_2(r) = \frac{2\Gamma(1+\beta)e^{r^2}}{\Gamma(1+2\beta)}$, $W_3(r) = 0$, we get

$$W_4(r) = \frac{4\Gamma(1 + \beta)\Gamma(1 + 3\beta)e^{r^2}}{\Gamma(1 + 2\beta)\Gamma(1 + 4\beta)}.$$

Continue the same procedure, we can get

$$W_5(r) = 0,$$

and

$$W_6(r) = \frac{8\Gamma(1 + \beta)\Gamma(1 + 3\beta)\Gamma(1 + 5\beta)e^{r^2}}{\Gamma(1 + 2\beta)\Gamma(1 + 4\beta)\Gamma(1 + 6\beta)}.$$

Now using the Taylor series $\tilde{\omega}(r, s) = \sum_{k=0}^{\infty} W_k(r)s^{k\beta}$, by putting the values we get the series solution as

$$\tilde{\omega}(r, s) = e^{r^2} \left(1 + \frac{2\Gamma(\beta + 1)s^{2\beta}}{\Gamma(2\beta + 1)} + \frac{2^2\Gamma(\beta + 1)\Gamma(3\beta + 1)s^{2(2\beta)}}{\Gamma(2\beta + 1)\Gamma(4\beta + 1)} + \frac{2^3\Gamma(\beta + 1)\Gamma(3\beta + 1)\Gamma(5\beta + 1)s^{2(3\beta)}}{\Gamma(2\beta + 1)\Gamma(4\beta + 1)\Gamma(6\beta + 1)} + \dots \right).$$

For $\beta = 1$, therefore series solution transferred to $\tilde{\omega}(r, s) = e^{r^2+s^2}$, which indicates an exact solution to CRDPDE. For problem 3, the results in Table 8 show the comparison of approximate solutions, while the results in Table 9 indicate the absolute error of approximate solutions obtained by RDTM. The comparison and absolute error of approximate solutions obtained by RDTM show the authentic accuracy of the proposed method.

Here, we declare some figures to show the quality of the technique (RDTM). From the results of Figure 7 provided that when β varies from 0.7 to 1 then RDTM solution converges to exact solution of problem 3, but Figure 8 shows RDTM solution for various values of β , whereas for $\beta = 1$ RDTM solution and exact solution are closed to each other see in Figure 9.

Table 8. Reduced differential transformed method (RDTM) solution at $\beta = 0.7, 0.8, 0.9$ and $r = 1$ by taking distinct values of s of problem 3

s	RDTM ($\beta = 0.7$)	RDTM ($\beta = 0.8$)	RDTM ($\beta = 0.9$)
0	2.718281828461	2.718281828461	2.718281828461
0.040	2.733123731161	2.728180110412	2.72485818441
0.080	2.766881283881	2.752933367941	2.742913878661
0.120	2.816201835882	2.790755173082	2.771805871542
0.160	2.88027427114	2.841212356141	2.811455374992
0.200	2.95912229771	2.904380429670	2.862045602260
0.240	3.05327635610	2.980667839410	2.923956067790
0.280	3.163664433550	3.070755513580	2.997741524441
0.320	3.291578671820	3.175578115280	3.084128277521
0.360	3.438679056610	3.2963263690	3.084128277521
0.400	3.6070209065611	3.434463126270	3.298510727050
0.440	3.799101582861	3.591750470	3.428902726690
0.480	4.017925669852	3.770287201920	3.576731358790

Table 9. Absolute error of the reduced differential transform method (RDTM) solution at $r = 1$ and corresponding to different values of s and taking $\beta = 1$ to the problem 3

s	Exact Solution ($\beta = 1$)	RDTM Solution ($\beta = 1$)	Error
0	2.71828182846	2.71828182846	0
0.040	2.72263456064	2.72263456064	$6.93889390391e^{-18}$
0.080	2.73573462153	2.73573462153	0
0.120	2.75770827592	2.75770827592	$6.93889390391e^{-18}$
0.160	2.78876821962	2.78876821962	$6.93889390391e^{-18}$
0.200	2.82921701435	2.82921701435	$6.93889390391e^{-18}$
0.240	2.879452005	2.879452005	$6.93889390391e^{-18}$
0.280	2.93997183096	2.93997183096	0
0.320	3.01138468133	3.01138468133	$6.93889390391e^{-18}$
0.360	3.09441848514	3.09441848514	0
0.400	3.18993327612	3.18993327612	$6.93889390391e^{-18}$
0.440	3.2989360256	3.2989360256	0
0.480	3.42259830184	3.42259830184	0

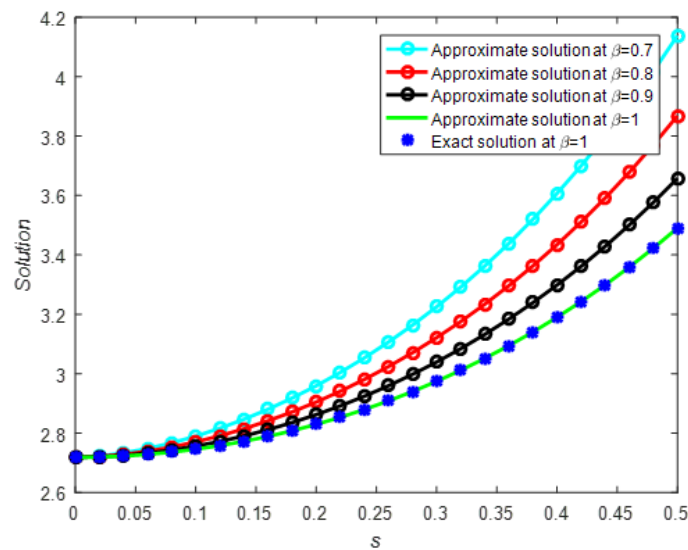


Figure 7. Exact and reduced differential transformed method (RDTM) results of problem 3 at $r = 1$ for some values of s and β

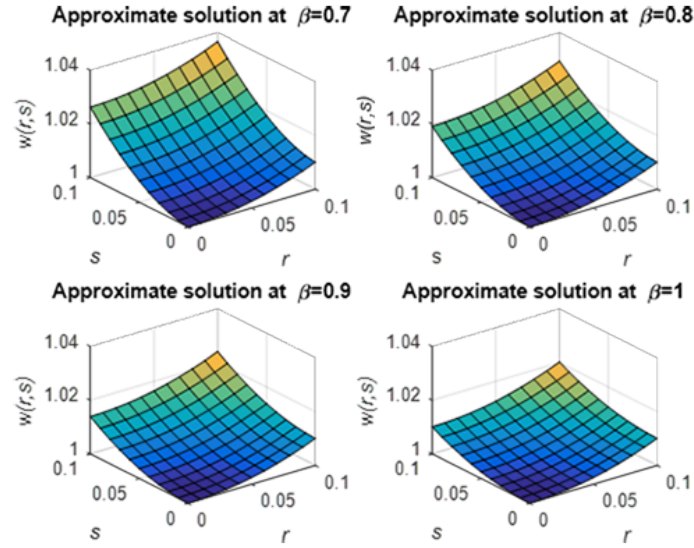


Figure 8. Plot of reduced differential transformed method (RDTM) results for different values of s and β

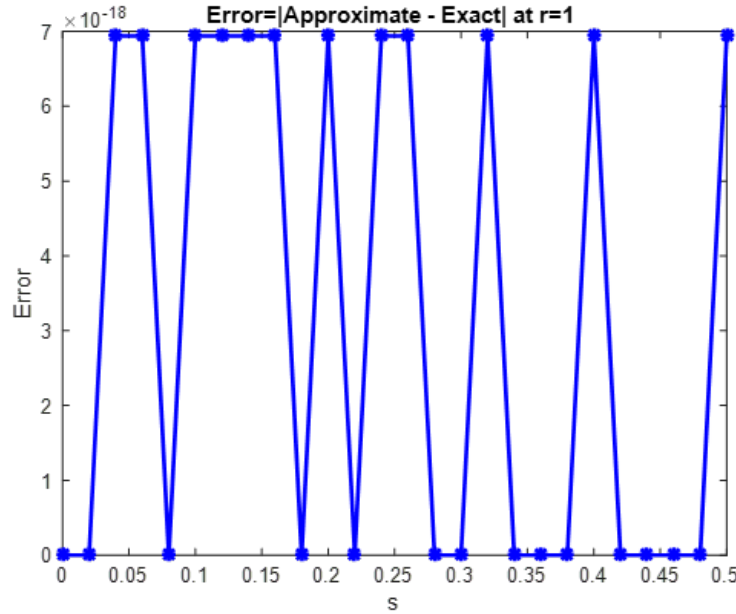


Figure 9. Error of reduced differential transformed method (RDTM) and exact results for different values of s and $\beta = 1$

4 Conclusions

In this manuscript, we have compared a powerful technique RDTM for the approximate solution of fractional order CRPDE and some of its special cases. The approximate solutions of the problems which are obtained by RDTM are very accurate and agree with exact solutions. The method is highly accurate and rapidly convergent that are useful for handling the physical problems that arise in various fields of biological science and engineering. By comparison of these results, we have observed that RDTM is a derived form of expansion of Taylor's series. Moreover, by observing the problem section, it is proposed that RDTM is more simple and easy in terms of calculation than other techniques. It can be used for the numerical treatment of coupled systems.

Author Contributions

Conceptualization, H.G. and S.M.; methodology, H.G.; software, H.G.; validation, H.G. and S.M.; formal analysis, H.G.; investigation, H.G. and S.M.; resources, H.G. and S.M.; data curation, H.G.; writing—original draft preparation, H.G. and S.M.; writing—review and editing, H.G. and S.M.; visualization, H.G.; supervision, H.G.; project administration, H.G. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data supporting the findings of this study are available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest.

References

- [1] H. Gul, H. Alrabaiah, S. Ali, K. Shah, and S. Muhammad, "Computation of solution to fractional order partial reaction diffusion equations," *J. Adv. Res.*, vol. 25, pp. 31–38, 2020. <https://doi.org/10.1016/j.jare.2020.04.021>
- [2] H. Gul, S. Ali, K. Shah, S. Muhammad, T. Sitthiwiratham, and S. Chasreechai, "Application of asymptotic homotopy perturbation method to fractional order partial differential equation," *Symmetry*, vol. 13, no. 11, p. 2215, 2021. <https://doi.org/10.3390/sym13112215>
- [3] B. Batiha, F. Ghanim, and K. Batiha, "Application of the new iterative method (NIM) to the generalized Burgers–Huxley equation," *Symmetry*, vol. 15, no. 3, p. 688, 2023. <https://doi.org/10.3390/sym15030688>
- [4] B. M. Yeolekar, R. D. Dave, and S. R. Khirsariya, "Solution of a cancer treatment model of a drug targeting treatment through nanotechnology using Adomian decomposition Laplace transform method," *Interactions*, vol. 245, no. 1, p. 278, 2024. <https://doi.org/10.1007/s10751-024-02114-6>
- [5] S. Kondo and T. Miura, "Reaction-diffusion model as a framework for understanding biological pattern formation," *Science*, vol. 329, no. 5999, pp. 1616–1620, 2010. <https://doi.org/10.1126/science.1179047>
- [6] S. Duran-Nebreda, J. Pla, B. Vidiella, J. Piñero, N. Conde-Pueyo, and R. Solé, "Synthetic lateral inhibition in periodic pattern forming microbial colonies," *ACS Synth. Biol.*, vol. 10, no. 2, pp. 277–285, 2021. <https://doi.org/10.1021/acssynbio.0c00318>
- [7] H. Meinhardt, *Models of Biological Pattern Formation*. London: Academic Press, 1982. https://www.researchgate.net/publication/281202174_Models_of_Biological_Pattern_Formation_Academic_Press_London_1982
- [8] J. D. Murray, *Mathematical Biology*. Berlin, Heidelberg: Springer, 1993. <https://doi.org/10.1007/978-3-662-08542-4>
- [9] A. Gupta and S. Chakraborty, "Linear stability analysis of high- and low-dimensional models for describing mixing-limited pattern formation in homogeneous autocatalytic reactors," *Chem. Eng. J.*, vol. 145, no. 3, pp. 399–411, 2009. <https://doi.org/10.1016/j.cej.2008.08.025>
- [10] J. D. Murray, E. A. Stanley, and D. L. Brown, "On the spatial spread of rabies among foxes," *Proc. R. Soc. Lond. B Biol. Sci.*, vol. 229, no. 1255, pp. 111–150, 1986. <https://doi.org/10.1098/rspb.1986.0078>
- [11] E. E. Holmes, M. A. Lewis, J. E. Banks, and R. R. Veit, "Partial differential equations in ecology: Spatial interactions and population dynamics," *Ecology*, vol. 75, no. 1, pp. 17–29, 1994. <https://doi.org/10.2307/1939378>
- [12] J. A. Sherratt and J. D. Murray, "Models of epidermal wound healing," *Proc. R. Soc. Lond. B Biol. Sci.*, vol. 241, no. 1300, pp. 29–36, 1990. <https://doi.org/10.1098/rspb.1990.0061>
- [13] M. A. J. Chaplain, "Reaction–diffusion prepatterning and its potential role in tumour invasion," *J. Biol. Syst.*, vol. 3, no. 4, pp. 929–936, 1995. <https://doi.org/10.1142/S0218339095000824>
- [14] J. A. Sherratt and M. A. Nowak, "Oncogenes, anti-oncogenes and the immune response to cancer: A mathematical model," *Proc. R. Soc. Lond. B Biol. Sci.*, vol. 248, no. 1323, pp. 261–271, 1992. <https://doi.org/10.1098/rspb.1992.0071>
- [15] R. A. Gatenby and E. T. Gawlinski, "A reaction-diffusion model of cancer invasion," *Cancer Res.*, vol. 56, no. 24, pp. 5745–5753, 1996.
- [16] S. Ali, K. Shah, and F. Jarad, "On stable iterative solutions for a class of boundary value problem of nonlinear fractional order differential equations," *Math. Methods Appl. Sci.*, vol. 42, no. 3, pp. 969–981, 2019. <https://doi.org/10.1002/mma.5407>
- [17] S. Bushnaq, S. Ali, K. Shah, and M. Arif, "Exact solution to non-linear biological population model with fractional order," *Therm. Sci.*, vol. 22, no. Suppl. 1, pp. S317–S327, 2018. <https://doi.org/10.2298/TSCI171127035B>
- [18] Y. Keskin and G. Oturanc, "Reduced differential transform method for partial differential equations," *Int. J. Nonlinear Sci. Numer. Simul.*, vol. 10, no. 6, pp. 741–750, 2009. <https://doi.org/10.1515/IJNSNS.2009.10.6.741>
- [19] H. Jafari, H. K. Jassim, S. P. Moshokoa, V. M. Ariyan, and F. Tchier, "Reduced differential transform method for partial differential equations within local fractional derivative operators," *Adv. Mech. Eng.*, vol. 8, no. 4, pp. 1–6, 2016. <https://doi.org/10.1177/1687814016633013>

- [20] O. Acan, M. M. Al Qurashi, and D. Baleanu, “Reduced differential transform method for solving time and space local fractional partial differential equations,” *J. Nonlinear Sci. Appl.*, vol. 10, no. 10, pp. 5230–5238, 2017. <https://doi.org/10.22436/jnsa.010.10.09>